

Chapter Three

FERMAT'S PRINCIPLE AND ITS APPLICATIONS

Now in the further development of science, we want more than just a formula. First we have an observation, then we have numbers that we measure, then we have a law which summarizes all the numbers. But the real glory of science is that we can find a way of thinking such that the law is evident. The first way of thinking that made the law about the behavior of light evident was discovered by Fermat in about 1650, and it is called the principle of least time, or Fermat's principle.

—Richard Feynman in *Feynman Lectures on Physics*, Vol. I

Learning Objectives

After reading this chapter, the reader should be able to:

- LO 1:** state Fermat's principle and its modified version.
- LO 2:** derive laws of reflection and refraction from Fermat's principle.
- LO 3:** describe various phenomena by solving the ray equation.
- LO 4:** obtain ray paths in inhomogeneous media.
- LO 5:** calculate the time taken by the rays in propagating through a parabolic index medium.

Important Milestones

- 140 AD** Greek physicist Claudius Ptolemy measured the angle of refraction in water for different angles of incidence in air and tabulated it.
- 1621** Although the above-mentioned numerical table was made in 140 AD, it was only in 1621 that Willebrord Snell, a Dutch mathematician, discovered the law of refraction which is now known as Snell's law.
- 1637** Descartes derived Snell's law; his derivation assumed corpuscular model of light.
- 1657** Pierre de Fermat enunciated his principle of 'least time' and derived Snell's law of refraction and showed that if the velocity of light in the second medium is less, the ray would bend towards the normal, contrary to what is predicted by the 'corpuscular theory'.

3.1 INTRODUCTION

LO 1

The study of the propagation of light in the realm of geometrical optics employs the concept of rays. To understand what a ray is, consider a circular aperture in front of a point source P as shown in Fig. 3.1. When the diameter of the aperture is quite large (~ 1 cm), then on the screen SS' , one can see a patch of light with well-defined boundaries. When we start decreasing the size of the aperture, then at first the size of the patch starts decreasing, but when the size

of the aperture becomes very small ($\lesssim 0.1$ mm) then the pattern obtained on SS' ceases to have well-defined boundaries. This phenomenon is known as diffraction and is a direct consequence of the finiteness of the wavelength (which is denoted by λ). In Chapters 18, 19 and 20 we will discuss the phenomenon of diffraction in great detail and will show that the diffraction effects become smaller with the decrease in wavelength and indeed in the limit of $\lambda \rightarrow 0$, the diffraction effects will be absent, and even for extremely small diameters of the aperture, we will obtain a well-defined

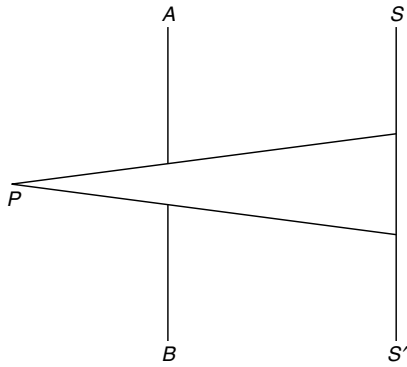


Fig. 3.1 The light emitted by the point source P is allowed to pass through a circular hole and if the diameter of the hole is very large compared to the wavelength of light, then the light patch on the screen SS' has well-defined boundaries.

shadow on the screen SS' . Therefore, in the zero wavelength limit one can obtain an infinitesimally thin pencil of light; this is called a *ray*. Thus, a ray defines the path of propagation of the energy in the limit of the wavelength going to zero. Since light has a wavelength of the order of 10^{-5} cm, which is small compared to the dimensions of normal optical instruments like lenses, mirrors, etc., one can, in many applications, neglect the finiteness of the wavelength. The field of optics under such an approximation (i.e., the neglect of the finiteness of the wavelength) is called geometrical optics.

The field of geometrical optics can be studied by using Fermat's principle which determines the path of the rays. According to this principle, *the ray will correspond to that path for which the time taken is an extremum in comparison to nearby paths*, i.e., it is either a minimum or a maximum or stationary*. Let $n(x, y, z)$ represent the position-dependent refractive index. Then

$$\frac{ds}{c/n} = \frac{n ds}{c}$$

will represent the time taken to traverse the geometric path ds in a medium of refractive index n . Here, c represents the speed of light in free space. Thus, if τ represents the total time taken by the ray to traverse the path AB along the curve C (see Fig. 3.2) then

$$\tau = \frac{1}{c} \sum_i n_i ds_i = \frac{1}{c} \int_{A \rightarrow B} n ds \quad (3.1)$$

where ds_i represents the i^{th} arc length and n_i the corresponding refractive index; the symbol $\int_{A \rightarrow B}$ below the integral represents the fact that the integration is from

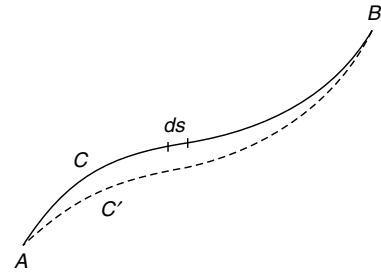


Fig. 3.2 If the path ACB represents the actual ray path then the time taken in traversing the path ACB will be an extremum in comparison to any nearby path $AC'B$.

the point A to B through the curve C . Let τ' be the time taken along the nearby path $AC'B$ (shown as the dashed curve in Fig. 3.2), and if ACB indeed represents the path of a ray, then τ will be either less than, greater than or equal to τ' for *all* nearby paths like $AC'B$. Thus according to Fermat's principle, out of the many paths connecting the two points, the light ray would follow that path for which the time taken is an extremum. Since c is a constant, one can alternatively define a ray as the path for which

$$\int_{A \rightarrow B} n ds \quad (3.2)$$

is an extremum**; the above integral represents the optical path from A to B along C ; i.e., the ray would follow the path for which

$$\delta \int_{A \rightarrow B} n ds = 0 \quad (3.3)$$

where the left-hand side represents the change in the value of the integral due to an infinitesimal variation of the ray path. We may mention here that according to the original statement of Fermat:

The actual path between two points taken by a beam of light is the one which is traversed in the least time.

The above statement is incomplete and slightly incorrect. The correct form is:

The actual ray path between two points is the one for which the optical path length is stationary with respect to variations of the path.

This is expressed by Eq. (3.3) and in this formulation, the ray paths may be maxima, minima or stationary.

From the above principle one can immediately see that in a homogenous medium (i.e., in a medium whose refractive

* The entire field of classical optics (both geometrical and physical) can be understood from Maxwell's equations, and of course, Fermat's principle can be derived from Maxwell's equations (see Refs. 3.1 and 3.2).

** A nice discussion on the extremum principle has been given in Chapter 26 of Ref. 3.3.

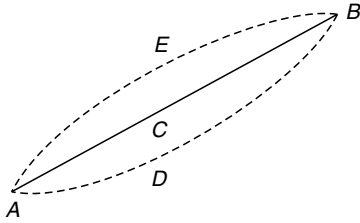


Fig. 3.3 Since the shortest distance between two points is along a straight line, light rays in a homogenous medium are straight lines; all nearby paths like AEB or ADB will take longer times.

index is constant at each point), the rays will be straight lines because a straight line will correspond to a minimum value of the optical path connecting two points in the medium. Thus referring to Fig. 3.3, if A and B are two points in a homogenous medium, then the ray path will be along the straight line ACB because any nearby path like ADB or AEB will correspond to a longer optical path.

3.2 LAWS OF REFLECTION AND REFRACTION FROM FERMAT'S PRINCIPLE

LO 2

To obtain the laws of reflection and refraction from Fermat's principle, consider a plane mirror MN as shown in Fig. 3.4. We have to determine the path from A to B (via the mirror) which has the minimum optical path length. Since the path would lie completely in a homogenous medium, we need to minimize only the path length. Thus we have to find that path APB for which $AP + PB$ is a minimum. To find the position of P on the mirror, we drop a perpendicular from A on the mirror

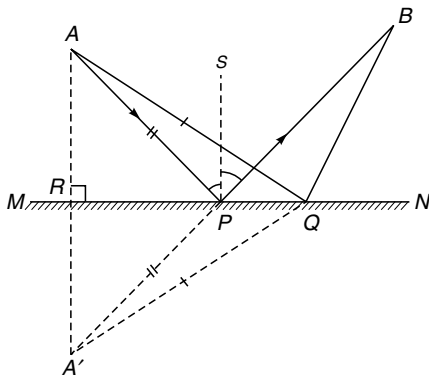


Fig. 3.4 The shortest path connecting the two points A and B via the mirror is along the path APB where the point P is such that AP, PS and PB are in the same plane and $\angle APS = \angle SPB$; PS being the normal to the plane of the mirror. The straight line path AB is also a ray.

and let A' be a point on the perpendicular such that $AR = RA'$; thus $AP = PA'$ and $AQ = A'Q$ where AQB is another path adjacent to APB . Thus we have to minimize the length $A'PB$. Clearly, for $A'PB$ to be a minimum, P must be on the straight line $A'B$. Thus, the points A, A' , P and B will be in the same plane and if we draw a normal PS at P then this normal will also lie in the same plane. Simple geometric considerations show that

$$\angle APS = \angle SPB$$

Thus for minimum optical path length, the angle of incidence $i (= \angle APS)$ and the angle of reflection $r (= \angle SPB)$ must be equal and the incident ray, the reflected ray and the normal to the surface at the point of incidence on the mirror must be in the same plane. These form the laws of reflection. It should be pointed out that, in the presence of the mirror there will be two ray paths which will connect the points A and B; the two paths will be AB and APB. Fermat's principle tells us that whenever the optical path length is an extremum, we will have a ray, and thus, in general, there may be more than one ray path connecting two points.

To obtain the laws of refraction, let PQ be a surface separating two media of refractive indices n_1 and n_2 as shown in Fig. 3.5. Let a ray starting from the point A, intersect the interface at R and proceed to B along RB. Clearly, for minimum optical path length, the incident ray, the refracted ray and the normal to the interface must all lie in the same plane. To determine that point R for which the optical path length from A to B is a minimum, we drop perpendiculars AM and BN from A and B respectively on the interface PQ. Let $AM = h_1$, $BN = h_2$ and $MR = x$. Then since A and B are fixed, $RN = L - x$, where $MN = L$ is a fixed quantity. The optical path length from A to B, by definition, is

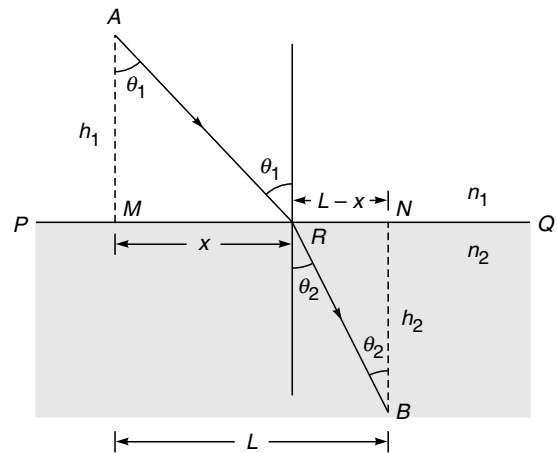


Fig. 3.5 A and B are two points in media of refractive indices n_1 and n_2 . The ray path connecting A and B will be such that $n_1 \sin \theta_1 = n_2 \sin \theta_2$.

$$\begin{aligned}
 L_{\text{op}} &= n_1 AR + n_2 RB \\
 &= n_1 \sqrt{x^2 + h_1^2} + n_2 \sqrt{(L-x)^2 + h_2^2}
 \end{aligned} \quad (3.4)$$

To minimize this, we must have

$$\frac{dL_{\text{op}}}{dx} = 0$$

$$\text{i.e., } \frac{n_1 x}{\sqrt{x^2 + h_1^2}} - \frac{n_2 (L-x)}{\sqrt{(L-x)^2 + h_2^2}} = 0 \quad (3.5)$$

Further, as can be seen from Fig. 3.5

$$\sin \theta_1 = \frac{x}{\sqrt{x^2 + h_1^2}}$$

$$\text{and } \sin \theta_2 = \frac{(L-x)}{\sqrt{(L-x)^2 + h_2^2}}$$

Thus, Eq. (3.5) becomes

$$n_1 \sin \theta_1 = n_2 \sin \theta_2 \quad (3.6)$$

which is the Snell's law of refraction.

The laws of reflection and refraction form the basic laws for tracing light rays through simple optical systems, like a system of lenses and mirrors, etc.

Example 3.1 Consider a set of rays, parallel to the axis, incident on a paraboloidal reflector (see Fig. 3.6). Show by using Fermat's principle, that all the rays will pass through the focus of the paraboloid; a paraboloid is obtained by rotating a parabola about its axis. This is the reason why a paraboloidal reflector is used to focus parallel rays from a distant source, like in radio astronomy (see Figs. 3.7 and 3.8).

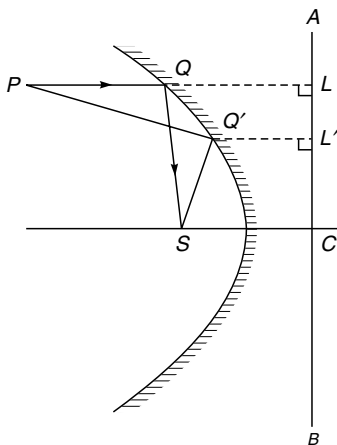


Fig. 3.6 All rays parallel to the axis of a paraboloidal reflector pass through the focus after reflection (the line ACB is the directrix). It is for this reason that antennas (for collecting electromagnetic waves) or solar collectors are often paraboloidal in shape.



Fig. 3.7 A paraboloidal satellite dish. [Photograph courtesy: McGraw-Hill Digital Access Library. A color photograph of the above figure appears as Fig. 1 in the prelim pages].



Fig. 3.8 Fully steerable 45 m paraboloidal dishes of the Giant Metrewave Radio Telescope (GMRT) in Pune, India. The GMRT consists of 30 dishes of 45 m diameter with 14 antennas in the Central Array. [Photograph courtesy: Professor Govind Swarup, GMRT, Pune. A color photograph of the above figure appears as Fig. 2 in the prelim pages].

Solution: Consider a ray PQ , parallel to the axis of the parabola, incident at the point Q (see Fig. 3.6). In order to find the reflected ray, one has to draw a normal at the point Q and then draw the reflected ray. It can be shown from geometrical considerations that the reflected ray QS will always pass through the focus S . However, this procedure will be quite cumbersome and as we will show, the use of Fermat's principle leads us to the desired results immediately.

To use Fermat's principle, we try to find out the ray connecting the focus S and an arbitrary point P (see Fig. 3.6). Let the ray path be $PQ'S$. According to Fermat's principle, the ray path will correspond to a minimum value of $PQ' + Q'S$. From the point Q' we drop a perpendicular $Q'L'$, on the directrix AB . From the definition of the parabola it follows that $Q'L' = Q'S$. Thus,

$$PQ' + Q'S = PQ + Q'L'$$

Let L be the foot of the perpendicular drawn from the point P on AB . Then, for $PQ' + Q'L'$ to be a minimum, the point Q should lie on the straight line PQL , and thus the actual ray which connects the points P and S will be $PQ + QS$ where PQ is parallel to the axis. Therefore, all rays parallel to the axis will pass through S and conversely, all rays emanating from the point S will become parallel to the axis after suffering a reflection.

Example 3.2 Consider an elliptical reflector whose foci are the points S_1 and S_2 (see Fig. 3.9). Show that all rays emanating from the point S_1 will pass through the point S_2 after undergoing a reflection.

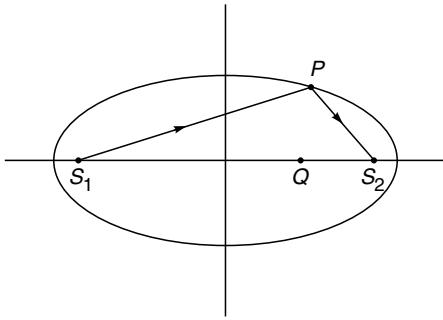


Fig. 3.9 All rays emanating from one of the foci of an ellipsoidal reflector will pass through the other focus.

Solution: Consider an arbitrary point P on the ellipse (see Fig. 3.9). It is well known that $S_1P + S_2P$ is a constant and therefore, all rays emanating from the point S_1 will pass through S_2 . (Notice that here we have an example where the time taken by the ray is stationary, i.e., it is neither a maximum nor a minimum but has a constant value for all points lying on the mirror.) As a corollary, we may note the following two points:

- Excepting the rays along the axis, no other ray (emanating from either of the foci) will pass through an arbitrary point Q which lies on the axis.
- The above considerations will remain valid even for an ellipsoid of revolution obtained by rotating the ellipse about its major axis.

Because of the above-mentioned property of elliptical reflectors, they are often used in laser systems. For example, in a ruby laser (see Sec. 27.3) one may have a configuration in which the laser rod and the flash lamp coincide with the focal lines of a cylindrical reflector of elliptical cross section; such a configuration leads to an efficient transfer of energy from the lamp to the ruby rod.

Example 3.3 Consider a spherical refracting surface SPM separating two media of refractive indices n_1 and n_2 (see Fig. 3.10). The point C represents the center of the spherical surface SPM . Consider two points O and Q such that the points O , C and Q are

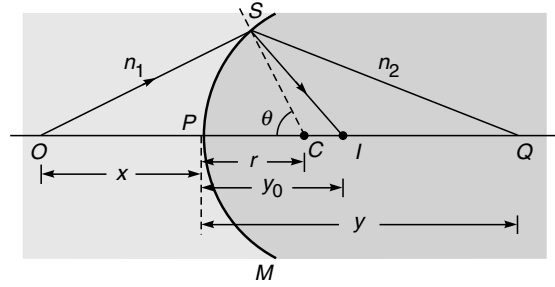


Fig. 3.10 SPM is a spherical refracting surface separating two media of refractive indices n_1 and n_2 . C represents the center of the spherical surface.

in a straight line. Calculate the optical path length OSQ in terms of the distances x , y , r and the angle θ (see Fig. 3.10). Use Fermat's principle to find the ray connecting the two points O and Q . Also, assuming the angle θ to be small, determine the paraxial image of the point O .

[Note: We reserve the symbol R to represent the radius of curvature of a spherical surface which will be positive (or negative) depending upon whether the center of curvature lies on the right (or left) of the point P . The quantity r represents the magnitude of the radius of curvature which, for Fig. 3.10, happens to be R . Similarly, the quantities x and y are the magnitudes of the distances; the sign convention is discussed later on in this problem.]

Solution: From the triangle SOC , we have

$$\begin{aligned} OS &= [(x+r)^2 + r^2 - 2(x+r)r \cos \theta]^{1/2} \\ &\approx \left[x^2 + 2rx + 2r^2 - 2(xr + r^2) \left(1 - \frac{\theta^2}{2} \right) \right]^{1/2} \\ &\approx x \left[1 + \frac{rx + r^2}{x^2} \theta^2 \right]^{1/2} \approx x + \frac{1}{2} r^2 \left(\frac{1}{r} + \frac{1}{x} \right) \theta^2 \end{aligned}$$

where we have assumed θ (measured in radians) to be small so that we may use the expression

$$\cos \theta \approx 1 - \frac{\theta^2}{2}$$

and also make a binomial expansion. Similarly, by considering the triangle SCQ , we would have

$$SQ \approx y - \frac{1}{2} r^2 \left(\frac{1}{r} - \frac{1}{y} \right) \theta^2$$

Thus the optical path length OSQ is given by

$$\begin{aligned} L_{op} &= n_1 OS + n_2 SQ \\ &\approx (n_1 x + n_2 y) + \frac{1}{2} r^2 \left[\frac{n_1}{x} + \frac{n_2}{y} - \frac{n_2 - n_1}{r} \right] \theta^2 \end{aligned} \quad (3.7)$$

For the optical path to be an extremum we must have

$$\frac{dL_{op}}{d\theta} = 0 = r^2 \left[\frac{n_1}{x} + \frac{n_2}{y} - \frac{n_2 - n_1}{r} \right] \theta \quad (3.8)$$

quantity and the image distance v and the radius of curvature R are positive quantities. For u to be positive, we must have a situation like the one shown in Fig. 4.1(b); in the absence of a refracting surface, the rays converge to a point to the right of P .

3. The angle that the ray makes with the axis is positive if the axis has to be rotated in the anticlockwise direction (through the acute angle) to coincide with the ray. Conversely, if the axis has to be rotated in the clockwise direction (through the acute angle) to coincide with the ray, then the slope angle is negative. Thus in Fig. 4.1(a), if θ_1 and θ_2 are the angles that the rays OS and SI make with the axis, then $\theta_1 = \alpha_1$ and $\theta_2 = -\alpha_2$; α_1 , α_2 and β represent the magnitudes of the angles. (If the final result does not depend on the angles, then it is more convenient to use the magnitude of the angles as has indeed been done in Sec. 4.2). In Fig. 4.1(b), both θ_1 and θ_2 are negative quantities.
4. The angle that a ray makes with the normal to the surface is positive if the normal has to be rotated in the anticlockwise direction (through the acute angle) to coincide with the ray, and conversely. Thus, in Fig. 4.1(a), ϕ_1 and ϕ_2 are positive quantities.
5. All distances measured upward from the axis (along a perpendicular to the axis) are positive, and all distances measured in the downward direction are negative.

4.2.2 The Gaussian Formula for a Single Spherical Surface

If we now use the sign convention discussed above, then for the ray diagram shown in Fig. 4.1(a), $u = -x$, $v = y$ and $R = r$. Thus Eq. (4.4) becomes

$$\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R} \quad (4.5)$$

which gives the image point due to refraction at a spherical surface (see also Eq. 3.10 of the previous chapter). Equation (4.5) is known as the **Gaussian formula for a single spherical surface**. It should be noted that, corresponding to Fig. 4.1(a), u is negative and v positive, whereas for Fig. 4.1(b) u and v are both positive.

Example 4.1 Consider a medium of refractive index 1.5 bounded by two spherical surfaces $S_1P_1M_1$ and $S_2P_2M_2$ as shown in Fig. 4.2. The radii of curvature of the two surfaces are 15 and 25 cm with their centers at C_1 and C_2 , respectively. There is an object at a distance of 40 cm (from P_1) on the line joining C_1 and C_2 . Determine the position of the paraxial image.

Solution: We first consider refraction by $S_1P_1M_1$. Obviously $u = -40$ cm, $R = +15$ cm, $n_1 = 1.0$ and $n_2 = 1.5$. Thus,

$$\frac{1.5}{v} + \frac{1}{40} = \frac{0.5}{15} \Rightarrow v = +180 \text{ cm}$$

In the absence of the second surface, the image is formed at O' at a distance of 180 cm from P_2 . O' now acts as a virtual object and since it is to the right of $S_2P_2M_2$ we have, while considering refraction by the second surface, $u = +180$ cm, $R = -25$ cm, $n_1 = 1.5$ and $n_2 = 1.0$. Thus,

$$\frac{1.0}{v} - \frac{1.5}{180} = +\frac{0.5}{25}$$

giving

$$v = +33\frac{1}{3} \text{ cm}$$

and a real image is formed on the right of P_2 at a distance of $33\frac{1}{3}$ cm.

We should point out that while considering refraction by a single surface (as in Fig. 4.1), the axis of the system is defined by the line joining the object point O and the center of curvature C ; thus *any* ray from the point O (like OS) will be in the plane containing the axis and the normal at the point S and consequently, the refracted ray will *always* intersect the axis. On the other hand, if there is second refracting surface (like in Fig. 4.2 or as in a lens) then the line joining the two

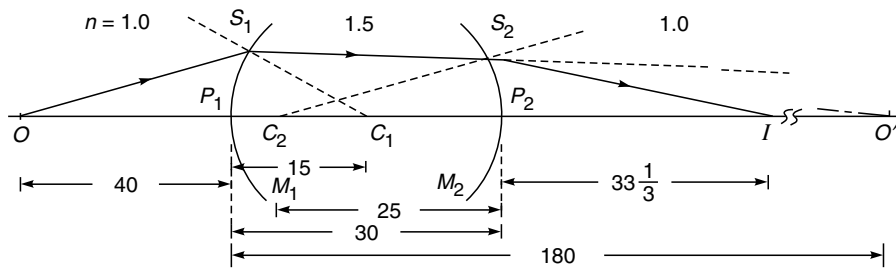


Fig. 4.2 Paraxial image formation by a medium of refractive index 1.5 bounded by two spherical surfaces $S_1P_1M_1$ and $S_2P_2M_2$. All distances are measured in centimeters.

centers of curvature is defined as the axis. In the latter case, as can be readily seen, not all rays from an off-axis point will intersect the axis and after refraction at the second surface will, in general, not remain confined to a single plane; these rays are known as *skew rays*. Rays which remain confined to a plane (containing the axis) are known as *meridional rays*; obviously, all rays emanating from a point on the axis are *meridional rays*.

4.3 REFLECTION BY A SINGLE SPHERICAL SURFACE

We next consider the imaging of a point object O by a spherical mirror SPM (see Fig. 4.3) in the paraxial approximation; the point C represents the center of curvature. We proceed in a manner exactly similar to that in Sec. 4.2 and we refer to Fig. 4.3 to obtain

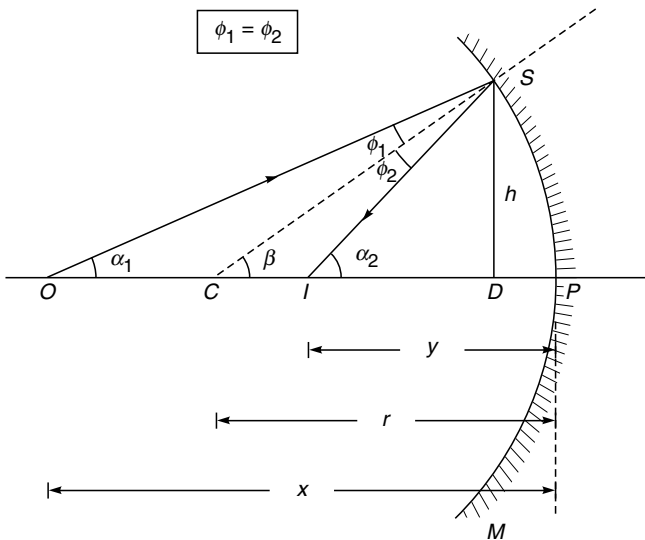


Fig. 4.3 Paraxial image formation by a spherical reflecting surface SPM .

$$\phi_1 = \beta - \alpha_1 \approx \frac{h}{r} - \frac{h}{x}$$

and

$$\phi_2 = \alpha_2 - \beta \approx \frac{h}{y} - \frac{h}{r}$$

where the distances x , y , h and r are defined in Fig. 4.3. Since $\phi_1 = \phi_2$ (the law of reflection), we get

$$\frac{1}{x} + \frac{1}{y} = \frac{2}{r} \quad (4.6)$$

If we again use the sign convention that all the distances to the right of P are positive and those to its left negative, then $u = -x$, $v = -y$ and $R = -r$; thus we obtain the mirror equation

$$\frac{1}{u} + \frac{1}{v} = \frac{2}{R} \quad (4.7)$$

which is the same as was derived by using Fermat's principle (see Problem 3.1). It is interesting to note that if we set $n_2 = -n_1$ in Eq. (4.5), we get Eq. (4.7). This follows from the fact that Snell's law of refraction Eq. (4.1) becomes the law of reflection if we have $n_2 = -n_1$.

We illustrate the use of Eq. (4.7) through an example.

Example 4.2 Consider an optical system consisting of a concave mirror $S_1P_1M_1$ and convex mirror $S_2P_2M_2$ of radii of curvatures 60 cm and 20 cm, respectively (see Fig. 4.4). We would like to determine the final image position of the object point O which is at a distance of 80 cm from the point P_1 , the two mirrors being separated by a distance of 40 cm.

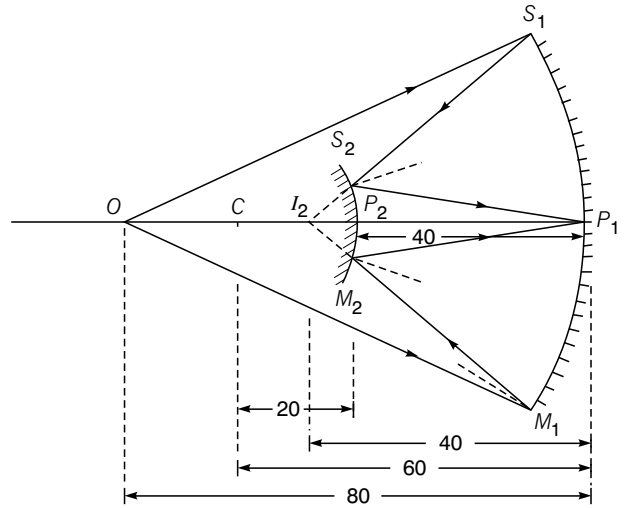


Fig. 4.4 Paraxial image formation by an optical system comprising a concave mirror $S_1P_1M_1$ and a convex mirror $S_2P_2M_2$.

We first consider the imaging by $S_1P_1M_1$; since $u = -80$ cm and $R = -60$ cm (because both O and C are on the left of P_1), we have

$$-\frac{1}{80} + \frac{1}{v} = -\frac{2}{60} \Rightarrow v = -48 \text{ cm}$$

In the absence of the mirror $S_2P_2M_2$, a real image will be formed at I_1 which now acts as a virtual object for $S_2P_2M_2$. Since I_1 is to the left of P_2 , we have (considering imaging by $S_2P_2M_2$), $u = -8$ cm and $R = -20$ cm, giving

$$\frac{1}{v} - \frac{1}{8} = -\frac{2}{20} \Rightarrow v = +40 \text{ cm}$$

Thus, the final image is formed on the right of $S_2P_2M_2$ at a distance of 40 cm, which happens to be the point P_1 .

4.4 THE THIN LENS

LO 2

A medium bounded by two spherical refracting surfaces is referred to as a spherical lens. If the thickness of such a lens (shown as t in Fig. 4.5) is very small compared to object and image distances, and to the radii of curvature of the refracting surfaces then the lens is referred to as a thin spherical lens. In general, a lens may have non-spherical refracting surfaces (e.g., it may have cylindrical surfaces). However, most lenses employed in optical systems have spherical refracting surfaces. Therefore, we will simply use the term ‘lens’ to imply a spherical lens. Different types of lenses are shown in Fig. 4.6. The line joining the centers of curvature of the spherical refracting surfaces is referred to as the *axis* of the lens.

In this section, we will consider the paraxial image formation by a thin lens. The corresponding considerations for a thick lens will be discussed in Problem 4.6.

We consider a point object O on the axis of a (thin) lens as shown in Fig. 4.5. The lens is placed in a medium of refractive index n_1 and the refractive index of the material of the lens is n_2 . Let R_1 and R_2 be the radii of curvature of the left and right surfaces of the lens; for the lens shown in Fig. 4.5, R_1 is positive and R_2 is negative. In order to determine the position of the image, we will consider successive refractions at the two surfaces; the image formed by the first surface is considered as the object (which may be real or virtual) for the second surface. Thus, if the second refracting surface had not been there, the image of the point O would have been formed at Q whose position (given by v') is determined from Eq. (4.8) (see also Eq. (4.5))

$$\frac{n_2}{v'} - \frac{n_1}{u} = \frac{n_2 - n_1}{R_1} \quad (4.8)$$

where u is the object distance which is negative for the object point O shown in the figure. Obviously if v' is

positive, then the point Q lies to the right of the surface and if v' is negative then Q lies to the left of the surface. The point Q now acts as the (virtual) object for the second refracting surface and the final image is formed at I whose position is determined from the equation

$$\frac{n_1}{v} - \frac{n_2}{v'} = \frac{n_1 - n_2}{R_2} \quad (4.9)$$

In Eqs. (4.8) and (4.9) the distances are measured from the center of the lens P ; this is justified because the lens has been assumed to be thin. Adding Eqs. (4.8) and (4.9), we get

$$\frac{1}{v} - \frac{1}{u} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \quad (4.10)$$

where

$$n = \frac{n_2}{n_1}$$

Equation (4.10) is known as the *thin lens formula* and is usually written in the form

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f} \quad (4.11)$$

where f , known as the focal length of the lens, is given by

$$\frac{1}{f} = (n - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \quad (4.12)$$

For a lens placed in air (which is usually the case), $n > 1$ and if $[(1/R_1) - (1/R_2)]$ is a positive quantity then the focal length is positive and the lens acts as a converging lens [see Fig. 4.7(a)]. Similarly, if $(1/R_1) - (1/R_2)$ is a negative quantity then the lens acts as a diverging lens (see Fig. 4.7(b)). However, if the double convex lens is placed in a medium whose refractive index is greater than that of the material of the lens, then the focal length becomes negative and the lens acts as a diverging lens [see Fig. 4.7(c)]; similarly for the double concave lens [see Fig. 4.7(d)].

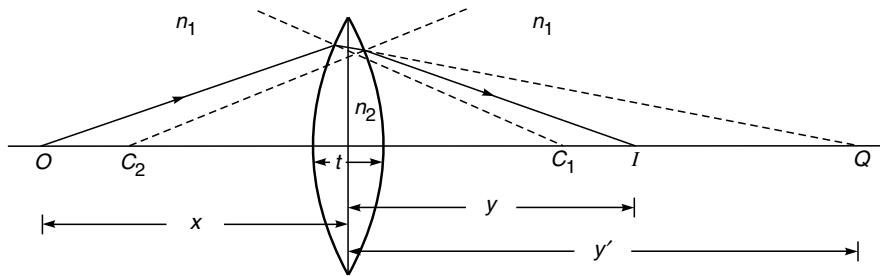


Fig. 4.5 Image formation by a thin lens. The line joining the two centers of curvature is known as the axis of the lens [$u = -x$, $v' = y'v = y$].

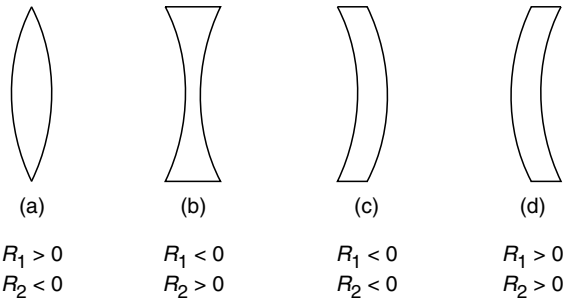


Fig. 4.6 Signs of R_1 and R_2 for different lens types.

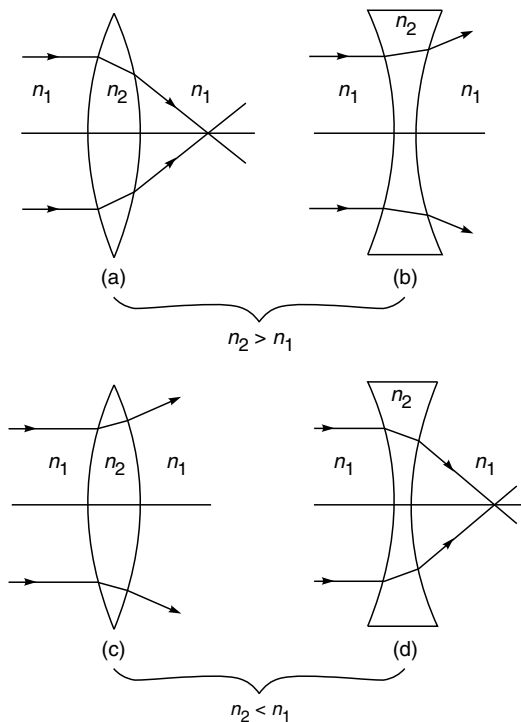


Fig. 4.7 (a) and (b) correspond to the situation when the refractive index of the material of the lens is greater than that of the surroundings and therefore a biconvex lens acts as a converging lens and a biconcave lens acts as a diverging lens. (c) and (d) correspond to the situation when the refractive index of the material of the lens is smaller than that of the surrounding medium and therefore a biconvex lens acts as a diverging lens and a biconcave lens as a converging lens.

4.5 THE PRINCIPAL FOCI AND FOCAL LENGTHS OF A LENS

For a converging lens, the *first principal focus* is defined as the point (on the axis) such that a ray passing through that

point will, after refraction through the lens, emerge parallel to the axis—see ray 1 in Fig. 4.8(a); the point F_1 is the first principal focus. For a diverging lens, the ray which (in the absence of the lens) would have passed through the first principal focus emerges, after refraction by the lens, as a ray parallel to the axis—see ray 1 in Fig. 4.8(b). The point F_1 is the **first principal focus** and its distance from the lens (denoted by f_1) is known as the **first focal length** of the lens. Obviously, f_1 is negative for a converging lens and positive for a diverging lens.

We next consider a ray which travels parallel to the axis [see ray 2 in Figs. 4.8(a) and (b)]. For a converging lens the point at which the ray will intersect the axis [shown as F_2 in Fig. 4.8(a)] is known as the **second principal focus** of the lens. Similarly, for a diverging lens, the point at which the ray would have intersected the axis (if produced backwards) is the second principal focus [see the point F_2 in Fig. 4.8(b)]. The distance of the second principal focus from the lens is known as the **second focal length** and is denoted by f_2 . As can be seen from Fig. 4.8, f_2 is positive for a converging lens and negative for a diverging lens.

For a thin lens placed in a medium such that the refractive indices on both sides of the lens are the same ($n_3 = n_1$ in Fig. 4.8), the values of f_1 and f_2 can be readily obtained by considering the thin lens formula [see Eq. (4.10)] and one gets

$$\frac{1}{f_2} = -\frac{1}{f_1} = \left[(n-1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \right] = \frac{1}{f} \quad (4.13)$$

However, if $n_3 \neq n_1$ then the thin lens formula assumes the form as given in Eq. (4.14) (see also Problem 4.2):

$$\frac{n_3}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R_1} + \frac{n_3 - n_2}{R_2} \quad (4.14)$$

Now, when $v = \infty$, $u = f_1$ (ray 1 in Fig. 4.8), we have

$$\frac{1}{f_1} = -\frac{1}{n_1} \left[\frac{n_2 - n_1}{R_1} + \frac{n_3 - n_2}{R_2} \right] \quad (4.15)$$

Similarly, when $u = -\infty$, $v = f_2$ (ray 2 in Fig. 4.8) we have

$$\frac{1}{f_2} = \frac{1}{n_3} \left[\frac{n_2 - n_1}{R_1} + \frac{n_3 - n_2}{R_2} \right] \quad (4.16)$$

Once we know f_1 and f_2 (and therefore the positions of the first and second principal foci) the (paraxial) image can be graphically constructed from the following rules:

- (i) A ray passing through the first principal focus will, after refraction, emerge parallel to the axis [see ray 1 in Figs. 4.8(a) and (b)]
- (ii) A ray parallel to the axis will, after refraction, either pass through or appear to come from (depending on the sign of f_2) the second principal focus [see ray 2 in Figs. 4.8(a) and (b)]
- (iii) A ray passing through the center of the lens P will pass through undeviated* [see ray 3 in Figs. 4.8(a) and (b)]

4.6 THE NEWTON FORMULA

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Let x_1 be the distance of the object from the first principal focus F_1 (x_1 will be positive if the object point is on the right of F_1 and conversely) and let x_2 be the distance of the image from the second principal focus F_2 as shown in Figs. 4.8(a) and (b). Considering similar triangles in Fig. 4.8(a), we have

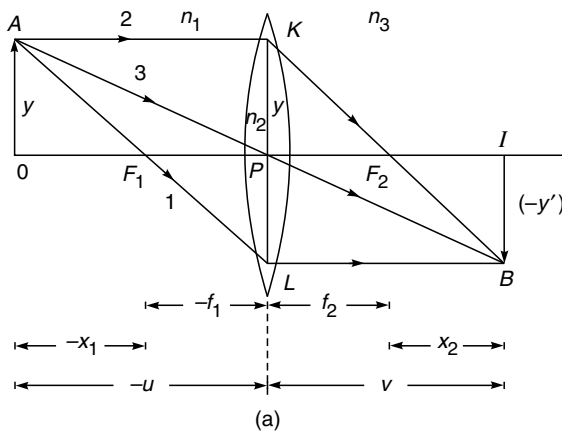
$$\frac{-y'}{y} = \frac{-f_1}{-x_1} \quad (4.17)$$

and

$$\frac{-y'}{y} = \frac{x_2}{f_2} \quad (4.18)$$

where the vertical distances are positive if measured above the line and negative if measured below the line (see Sec. 4.2.1). Equations (4.17) and (4.18) give

$$f_1 f_2 = x_1 x_2 \quad (4.19)$$



which is known as the *Newtonian lens formula*. It may be noted that for a diverging lens [see Fig. 4.8(b)], Eqs. (4.17) and (4.18) would be

$$\frac{y'}{y} = \frac{f_1}{-x_1} = \frac{x_2}{-f_2}$$

which are identical to Eqs. (4.17) and (4.18).

When the thin lens has the same medium on the two sides, then using Eq. (4.13) we have

$$x_1 x_2 = -f^2 \quad (4.20)$$

showing that x_1 and x_2 must be of opposite signs. Thus, if the object lies on the left of the first principal focus, then the image will lie on the right of the second principal focus, and vice versa.

4.7 LATERAL MAGNIFICATION

The lateral magnification m is the ratio of the height of the image to that of the object. Considering either Fig. 4.8(a) or (b) we readily get

$$m = \frac{y'}{y} = \frac{v}{u} = \frac{f_2 + x_2}{f_1 + x_1} = -\frac{f_1}{x_1} = -\frac{x_2}{f_2} \quad (4.21)$$

where we have made use of Eqs. (4.17) and (4.18). Obviously, if m is positive, the image is erect [as in Fig. 4.8(b) and conversely if m is negative, the image is inverted as in Fig. 4.8(a)].

The magnification can also be calculated as the product of the individual magnifications produced by each of the

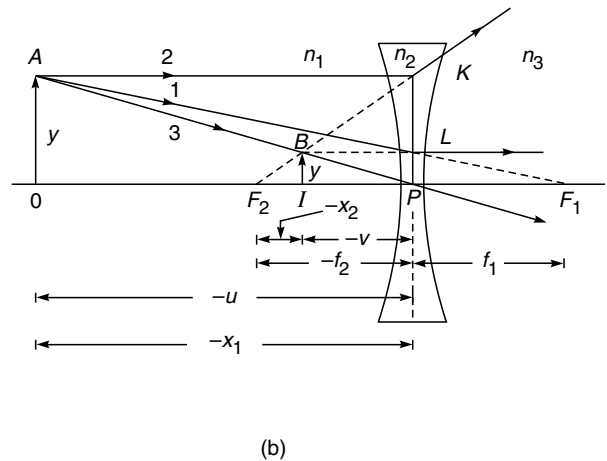


Fig. 4.8 (a) Paraxial imaging by a converging lens; x_1, f_1 and u are negative quantities and x_2, f_2 , and v are positive quantities. (b) Paraxial imaging by a diverging lens; here x_1, f_2, u and v are negative quantities and x_2 and f_1 are positive quantities.

* This follows from the fact that, for a thin lens, when $u = 0$, v is also equal to zero [see Eqs. (4.10) and (4.14)].

refracting surfaces; referring to Fig. 4.9, the magnification produced by a single refracting surface is given by

$$m = \frac{y'}{y}$$

and considering triangles AOC and ICB , we get

$$\frac{-y'}{y} = \frac{v - R}{-u + R} = \frac{\frac{v}{R} - 1}{-\frac{u}{R} + 1} \quad (4.22)$$

Now, Eq. (4.5) gives us

$$\frac{n_2}{n_1} - \frac{v}{u} = \frac{n_2 - n_1}{n_1} \frac{v}{R}$$

and

$$\frac{u}{v} - \frac{n_1}{n_2} = \frac{n_2 - n_1}{n_2} \frac{u}{R}$$

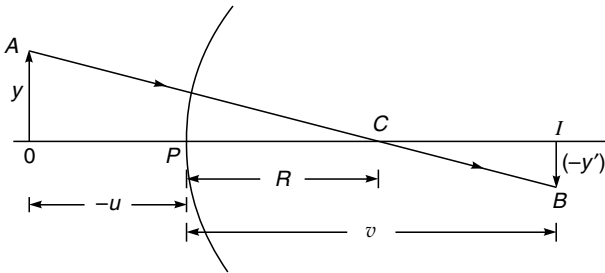


Fig. 4.9 Imaging of an object of height y by a spherical refracting surface.

Substituting for v/R and u/R in Eq. (4.22), we get

$$m = \frac{y'}{y} = \frac{n_1 v}{n_2 u} \quad (4.23)$$

Thus, if m_1 and m_2 represent the magnifications produced by the two refracting surfaces in Fig. 4.8, then

and

$$m = \frac{n_1}{n_2} \frac{v'}{u}$$

where v' represents the distance of the image formed by the first refracting surface. Thus,

$$m = m_1 m_2 = \frac{v}{u} \quad (4.24)$$

consistent with Eq. (4.21).

Example 4.3 Consider a system of two thin lenses as shown in Fig. 4.10. The convex lens has a focal length of +20 cm and the concave lens has a focal length of -10 cm. The two lenses are separated by 8 cm. For an object of height 1 cm (at a distance of 40 cm from the convex lens), calculate the position and size of the image. (The same problem will be solved again in Chapter 5 by using the matrix method.)

Solution: Let us first calculate the position and size of the image formed by the first lens:

$$u = -40 \text{ cm}, f = +20 \text{ cm}$$

Therefore, using Eq. (4.11), we get

$$\frac{1}{v} = \frac{1}{u} + \frac{1}{f} = -\frac{1}{40} + \frac{1}{20} = +\frac{1}{40}$$

Thus, $v = +40$ cm and $m_1 = -1$; the image is of the same size but inverted. This image acts as a virtual object for the concave lens with $u = +32$ cm and $f = -10$ cm. Thus,

$$\frac{1}{v} = \frac{1}{32} - \frac{1}{10} = -\frac{22}{320}$$

$$v \approx -14.5 \text{ cm}$$

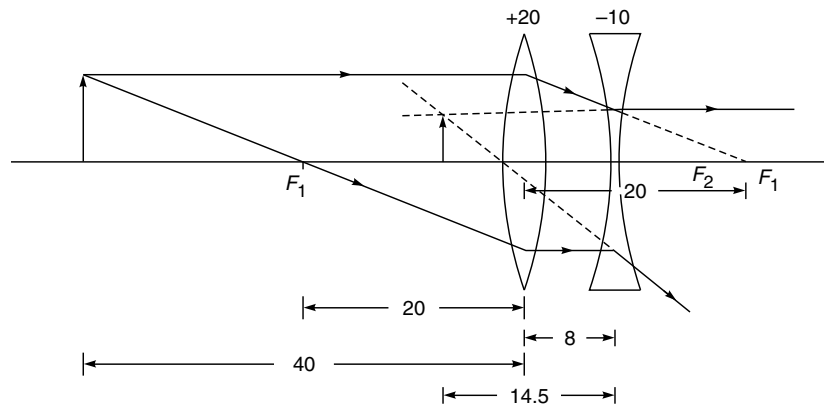


Fig. 4.10 Paraxial imaging by an optical system consisting of a converging lens of focal length 20 cm and a diverging lens of focal length -10 cm separated by 8 cm. All distances in the figure are in centimeters.